

Evaluation Report & Next-Step Execution Roadmap

Intern Name: N.S RAKSHNA

Assigned Construct: RIASEC Career Interests

Proposed Task: Day-in-the-Life Project Simulation

Evaluation Status: Proceed with Modifications and Multi-World Expansion

1. Executive Summary

The proposed Day-in-the-Life Project Simulation demonstrates exceptional alignment with our vision of building an AI-driven, behavior-based career assessment platform. Instead of relying solely on self-report questionnaires, the simulation captures occupational preferences through actual choices, engagement patterns, and behavioral indicators. The concept is highly engaging, difficult to fake, and capable of generating rich behavioral analytics. However, the current “Tech Startup” theme may unintentionally favor Investigative, Enterprising, and Realistic preferences while underrepresenting Artistic, Social, and Conventional interests. To align with our startup objectives, the simulation should be redesigned into multiple worlds where all six RIASEC dimensions are equally represented.

2. Alignment with MentiScope's Objectives

- Alignment with Holland's RIASEC Theory: 9.5/10
- Scientific Validity: 9/10
- Innovation: 9.5/10
- Adaptive Potential: 9.5/10
- AI Readiness: 9.5/10
- MVP Feasibility: 9/10
- Overall Evaluation: 9.5/10 – Proceed with Modifications

3. Scientific Assessment

Primary dimensions to measure:

- Realistic (R)
- Investigative (I)
- Artistic (A)
- Social (S)
- Enterprising (E)
- Conventional (C)

Potential construct contamination:

- Domain familiarity (e.g., technology exposure)
- Prior experiences and socioeconomic opportunities
- Gaming strategies and reward-seeking behavior
- General cognitive abilities and decision-making speed

The task should primarily measure preference patterns, engagement tendencies, and work-environment interests rather than performance or academic knowledge.

4. Recommended Task Architecture

Module 1 – Multi-World Exploration

Introduce several environments such as Hospital, NGO, Film Production, Space Mission, Startup, Smart City, and Research Lab.

Measures: Initial attraction and curiosity patterns.

Module 2 – Activity Selection

Students choose tasks and responsibilities within each world.

Measures: Preference frequency and domain interests.

Module 3 – Persistence and Revisit Behaviour

Students decide whether to continue, switch, or revisit activities.

Measures: Engagement, commitment, and preference stability.

Module 4 – Adaptive Simulation

Present new challenges and role variations.

Measures: Preference consistency across changing contexts.

5. Recommended Scoring Model

- Choice Frequency Score – 30%
- Persistence Score – 20%
- Revisit Behaviour Score – 15%
- Selection Latency Score – 15%
- Cross-Context Consistency Score – 20%

6. AI Analytics to Capture

- Choice frequency by RIASEC dimension
- Selection latency
- Revisit behaviour
- Task persistence
- Abandonment rate

- Preference shifts over time
- Exploration diversity index
- Consistency across scenarios
- Engagement patterns
- Top three occupational interest profiles

7. MVP Deliverables (Current Phase)

- ✓ Activity-to-RIASEC mapping table
- ✓ Minimum 30 activities per RIASEC dimension
- ✓ Multi-world scenario bank
- ✓ Behavioral scoring engine
- ✓ Explainable recommendation logic
- ✓ Event logger
- ✓ Analytics dashboard
- ✓ Documentation
- ✓ Pilot dataset

8. Assessment SDK Compliance Requirements

All outputs must follow the common Assessment SDK specification.

Session Data:

student_id, session_id, construct, task_id, module_version, difficulty_level, start_time, end_time, timestamp.

Event Data:

event_id, student_id, session_id, construct, task_id, item_id, event_type, response, correct, reaction_time_ms, error_type, difficulty_level, timestamp.

Score Output:

raw_score, normalized_score, percentile, sub_scores, confidence_score.

Analytics Output:

behavioral_metrics and recommendations.

9. Next Sprint Execution Plan (2 Weeks)

Sprint Objective:

Redesign the simulation into balanced multi-world environments and develop a foundational MVP prototype.

Expected Deliverables:

1. Literature review on Holland's RIASEC theory and behavioral career assessments.
2. Construct map defining indicators for all six RIASEC dimensions.
3. Multi-world scenario specifications.
4. Activity-to-RIASEC mapping table.
5. Behavioral scoring engine design.
6. Explainable recommendation logic.
7. Event logging implementation.
8. Demonstration prototype of one world simulation.

10. Definition of Done (DoD)

Scientific:

- ✓ All six RIASEC dimensions equally represented.
- ✓ Bias toward specific environments minimized.
- ✓ Behavioral indicators clearly defined.

Technical:

- ✓ Multi-world scenario engine implemented.
- ✓ Behavioral scoring engine functional.
- ✓ Event logging and analytics dashboard available.
- ✓ Results exported in SDK format.

Validation:

- ✓ Internal consistency ≥ 0.80 .
- ✓ Pilot study with at least 100 students.
- ✓ Preliminary evidence of occupational preference measurement.
- ✓ Explainable and interpretable recommendations.

Guidance to the Intern

Your responsibility is not to build the entire platform at this stage. Your responsibility is to build a scientifically valid micro-assessment engine for RIASEC career interests that follows the common Assessment SDK. Focus on creating balanced environments, capturing authentic behavioral preferences, and generating explainable recommendations. This module will later be integrated into the unified cognitive and career assessment platform.